Supplementary Material

Neighbours and Orientation: Two Notions of In-Distribution Conformity in Frozen Diffusion Features

Anonymous Submission

S1 Protocol and Data

Splits. For every ID dataset the images are split into three disjoint parts. (i) *Selection data*: ID development images and OOD datasets that are never used for evaluation (for the five 32×32 ID datasets: STL-10, GTSRB, Flowers-102, EuroSAT, MNIST in RGB, Oxford-IIIT Pets, Caltech-101, Food-101, FGVC-Aircraft, Imagenette and Rendered SST-2; for ImageNet-64 the same without Imagenette; for FashionMNIST: EMNIST letters, Kuzushiji-49, Semeion and greyscale CIFAR-10, SVHN and GTSRB). (ii) *Development evaluation images*: 1,000 per set (2,000 for FashionMNIST). (iii) *Final evaluation images*: 500 per set. The configuration of Upright kNN (block set and R of the kNN term) was chosen on (i) among 20 candidates (half-resolution block or the mean of the three blocks; R among ten values from 1 to 1000) by the mean AUROC of the term alone; the head (inputs, linear vs. MLP, temperature) was chosen on held-out bank images only. The method was developed on (ii), a written protocol was frozen, and (iii) was evaluated once. All tables of the paper are on (iii) except the development ablations (Table S7) and the mirror test (Table S8).

OOD sets. For the 32×32 ID datasets: the other natural datasets among CIFAR-10/100, SVHN and CelebA, plus Textures (DTD), Tiny-ImageNet (validation images resized to 32×32), LSUN and ImageNet (the resized versions distributed with CSI), ImageNet32 (downsampled validation images) and Places365 (validation images). For ImageNet-64: the OpenOOD v1.5 sets (SSB-hard, NINCO, iNaturalist, Textures, OpenImage-O) plus Places365 and CelebA. For FashionMNIST: MNIST, KMNIST, Omniglot and notMNIST embedded in 32×32. Every ID dataset is also paired with its own images flipped upside down and, except ImageNet-64, with uniform-noise and constant images, giving 63 pairs.

Bank and noise. The bank is 50,000 ID training images (10,000 for ImageNet-64, 10 per class). The noise image ε of $\sigma_1\varepsilon$ is one fixed draw per noise seed, shared by all images; the results are averaged over three seeds.

S2 Models and Configuration

ID	Res.	Model	Source	d	R	T	Head acc. (%)	tuned kNN configuration
CIFAR-10	32×32	EDM (VP, DDPM++)	[Karras *et al.*, 2022]	19.8	2	1.43	95.5	mean of 3 blocks, σ_1, mean of top 5
SVHN	32×32	VP-SDE	[Kamkari *et al.*, 2024]	22.8	5	1.43	86.7	mean of 3 blocks, σ_1, mean of top 10
CelebA	32×32	iDDPM	[Nichol and Dhariwal, 2021]	25.1	5	0.72	100.0	mean of 3 blocks, σ_7, mean of top 5
CIFAR-100	32×32	DDPM++ (EDM precond.)	[Chen *et al.*, 2024]	24.0	5	1.38	88.2	mean of 3 blocks, σ_1, mean of top 10
ImageNet-64	64×64	EDM2-XS (uncond.)	[Karras *et al.*, 2024]	20.2	1000	1.18	77.3	mean of 3 blocks, σ_1, 1000th neighbour
FashionMNIST	32×32	U-Net (EDM precond.)	trained by us	14.4	1	0.97	99.7	half-res. block, σ_7, mean of top 2

Table S1: ID datasets, diffusion models and the configuration of Upright kNN (noise seed 0). All weights except FashionMNIST are public and unchanged. d: Levina–Bickel intrinsic dimension of the bank features (20 neighbours, 3,000 bank points); R: neighbour rank chosen on the selection data; T: temperature of the head; head accuracy: rotation accuracy on the 5,000 held-out bank images. The last column gives the configuration of the tuned kNN baseline (chosen on the selection data among 80).

S3 Comparison Methods

All likelihood- and statistic-based methods use the same diffusion models. NLL, IC and TT use the probability-flow ODE with 64 Heun steps (128 forward and 128 backward evaluations). MSMA* is the Mahalanobis distance of the vector of score norms at 20 noise levels, fitted on ID development images. DiffPath and GEPC follow the official code and EigenScore the paper's procedure. CFS follows the official code (hook policy: up to four encoder and four decoder candidates in structural order, the one with the largest variance on 32 ID training images selected in each region; channel mean and standard deviation; diagonal-Gaussian and standardised 10-NN heads fitted on the bank). At the paper's default noise level (logSNR 5) our port reproduces the paper's CIFAR-10 numbers on the four shared pairs within 0.5 points; the official configuration (logSNR 7) is better on the development images and is used throughout. The rotation classifier is a ResNet-18 trained from scratch for 30 epochs on the ID training images; its score is the sum of the log-probabilities of the correct angle over the four rotated inputs. CSI is described in Section S12. CPU times use a randomly initialised CIFAR-10 network (the cost does not depend on the weights) on an Intel Xeon (2.1 GHz, 2 threads).

S4 Two Notions: Details and Robustness

Analysis. All quantities of Table 1 of the paper are computed from the stored features of the frozen method (no change to it). *Geometry*: in the space of the kNN term (unit-length channel mean and standard deviation of the three blocks), the distance between each of 4,000 random bank images and its own half turn, divided by its distance to its R-th nearest other bank image (R of the frozen protocol); median. *Discriminant and principal directions*: in the head's input space (whole-image and 2×2 grid statistics of the three blocks, standardised on the fitting images), a linear discriminant analysis (LDA) of the four rotations of the bank images (within-class scatter regularised by a ridge λ), the top principal directions of all rotated views, and 20 draws of random orthonormal directions; we report the variance carried by three directions relative to three random directions. *Projected kNN*: the 10th-nearest-neighbour distance (Euclidean; cosine as a variant) to 10,000 bank images in the projection onto those directions, as a score for ID test images against upside-down ID images and against every natural OOD set. Table 1 uses the fit of the first analysis we ran (4,000 bank images, $\lambda = 10^{-3}$) on the final evaluation images, averaged over the three noise seeds; the largest standard deviation over seeds of a frame-3 natural-OOD mean is 0.6 points.

	LDA fit (final evaluation images, seed 0): var. frame / frame-3 flip / frame-3 natural												development images (seeds 0–2)			
	4k, $\lambda = 10^{-3}$			4k, $\lambda = 10^{-1}$			10k, $\lambda = 10^{-3}$			10k, $\lambda = 10^{-1}$			$\|\Delta_{180}\|/r_R$	fr.-3 flip	fr.-3 nat.	PC-3 nat.
CIFAR-10	0.17	94.5	84.6	2.9	96.5	88.1	0.31	97.6	89.1	4.0	97.7	89.7	1.11	94.1	84.7	75.2
SVHN	0.14	60.3	93.4	3.8	61.9	95.2	0.20	64.3	96.4	4.4	65.3	96.9	1.06	59.7	93.6	89.7
CelebA	1.7	100.0	100.0	20	100.0	100.0	2.3	100.0	100.0	22	100.0	100.0	1.71	100.0	100.0	100.0
CIFAR-100	0.08	74.9	63.1	1.3	78.7	59.8	0.13	80.6	62.0	1.7	80.6	59.6	0.81	75.2	62.4	65.3
ImageNet-64	0.06	78.8	55.6	1.3	83.1	53.5	0.11	87.8	57.3	1.9	86.1	52.2	0.30	78.7	55.3	80.3
FMNIST	0.39	99.9	99.6	7.5	99.9	99.5	0.49	100.0	99.6	8.2	99.9	99.6	2.55	99.9	99.3	93.9

Table S2: Robustness of the two-notions analysis. Left: the LDA fitted on 4,000 or 10,000 bank images with ridge 10^{-3} or 10^{-1}: variance of the three discriminant directions relative to three random directions, and AUROC (%) of the frame-3 kNN on upside-down ID images and on natural OOD sets (mean). Right: the same analysis on the development evaluation images.

Natural OOD sets in the projections. The paper uses the projected kNN rules only for upside-down images. On natural OOD sets the result depends on the distance (Table S3): with the cosine distance, used in our first (development) analysis, the three discriminant directions beat the top three principal directions on all six ID datasets; with the Euclidean distance they do so on four, and on ImageNet-64 they are at the level of random directions. Either way they do not exceed the orientation term, which reads the frame information out better than a distance in a linear projection does.

ID dataset	frame-3 Eucl.	frame-3 cos.	PC-3 Eucl.	PC-3 cos.	random 3 Eucl.	random 3 cos.	kNN term	orient. term
CIFAR-10	84.3	89.3	73.8	75.2	55.4	60.1	93.7	93.8
SVHN	93.1	94.3	90.3	85.6	71.7	58.2	99.4	95.7
CelebA	100.0	100.0	100.0	99.9	78.3	78.9	100.0	100.0
CIFAR-100	63.0	68.7	66.5	61.7	58.3	53.5	85.0	72.9
ImageNet-64	55.1	69.5	80.6	64.9	58.5	52.4	77.0	77.4
FMNIST	99.6	99.5	93.2	90.6	66.9	70.8	99.9	99.8

Table S3: Mean AUROC (%) over the natural OOD sets of kNN rules restricted to three directions (final evaluation images, seeds 0–2, fit of Table 1 of the paper; random: mean of 20 draws), with the two terms of Upright kNN for reference.

Table S4: The two terms per pair (final evaluation images, three seeds): AUROC (%) of the kNN term, the orientation term and their sum (Upright kNN); fraction (%) of the OOD images flagged by the kNN term only, by the orientation term only and by both, at the thresholds that keep 95% of the ID images; Spearman correlation ρ of the two terms on the OOD images.

ID	OOD	kNN term	orient. term	sum	kNN only	orient. only	both	ρ
CIFAR-10	CIFAR-100	77.4	86.0	80.3	13.0	20.2	19.0	0.42
	SVHN	99.7	99.3	99.9	0.3	0.3	99.4	-0.17
	CelebA	89.9	97.2	96.0	2.9	62.7	22.6	0.10
	Textures	99.5	98.0	99.5	4.9	2.0	92.2	0.18
	Tiny-ImageNet	92.0	94.3	96.1	10.5	39.0	36.7	0.11
	LSUN	96.0	92.9	96.7	29.9	12.6	45.1	0.09
	ImageNet	96.9	94.5	97.6	19.8	10.3	61.2	0.25
	ImageNet32	96.0	90.4	96.3	26.2	8.8	50.5	0.36
	Places365	95.5	91.0	96.0	28.3	11.3	43.9	0.28
	upside-down ID	77.8	99.2	96.1	0.5	78.1	19.0	0.16
SVHN	CIFAR-10	99.1	96.7	99.6	15.0	1.4	82.3	0.44
	CIFAR-100	99.3	95.6	99.7	21.9	0.4	76.7	0.33
	CelebA	98.2	93.0	99.2	34.7	2.3	59.1	0.16
	Textures	99.8	93.8	99.9	31.2	0.0	68.5	0.18
	Tiny-ImageNet	99.3	95.0	99.7	24.8	0.9	73.8	0.42
	LSUN	99.7	97.1	99.9	15.7	0.5	83.5	0.47
	ImageNet	99.7	96.5	99.9	16.6	0.5	82.6	0.37
	ImageNet32	99.7	97.3	99.9	14.5	0.2	85.0	0.39
	Places365	99.7	96.7	99.8	16.4	0.3	82.9	0.52
	upside-down ID	56.1	78.6	64.5	4.7	25.1	2.6	0.12
CelebA	CIFAR-10	100.0	100.0	100.0	0.0	0.0	100.0	0.10
	CIFAR-100	100.0	100.0	100.0	0.0	0.0	100.0	0.13
	SVHN	100.0	100.0	100.0	0.0	0.0	100.0	0.15
	Textures	100.0	100.0	100.0	0.0	0.0	100.0	-0.08
	Tiny-ImageNet	99.9	100.0	100.0	0.0	0.4	99.6	0.10
	LSUN	99.9	100.0	100.0	0.0	0.3	99.7	0.14
	ImageNet	100.0	100.0	100.0	0.0	0.0	100.0	0.19
	ImageNet32	100.0	100.0	100.0	0.0	0.0	100.0	0.09
	Places365	100.0	100.0	100.0	0.0	0.2	99.8	0.24
	upside-down ID	99.5	100.0	100.0	0.0	0.0	100.0	0.23
CIFAR-100	CIFAR-10	66.3	28.4	64.1	12.0	0.6	0.7	0.29
	SVHN	98.8	95.1	99.5	34.7	4.1	59.8	-0.10
	CelebA	77.4	73.8	78.4	2.2	11.1	0.5	0.20
	Textures	98.7	90.7	99.0	55.5	2.4	36.7	0.03
	Tiny-ImageNet	82.2	82.3	86.4	13.5	27.0	5.1	-0.07
	LSUN	81.8	67.9	81.7	9.3	8.8	0.6	-0.06
	ImageNet	87.7	76.7	88.3	20.7	14.4	6.7	0.21
	ImageNet32	89.0	73.7	88.9	29.1	7.9	8.5	0.22

ID	OOD	kNN term	orient. term	sum	kNN only	orient. only	both	ρ
	Places365	83.2	67.1	83.0	13.5	9.3	4.1	0.17
	upside-down ID	56.5	95.0	79.1	1.6	69.4	3.5	-0.01
ImageNet-64	SSB-hard	67.8	58.7	67.3	14.3	4.5	0.9	0.48
	NINCO	68.3	72.1	71.7	8.9	13.7	0.6	0.06
	iNaturalist	68.1	75.1	71.9	1.9	14.8	0.7	0.05
	Textures	91.0	87.9	93.6	29.7	19.4	14.5	0.01
	OpenImage-O	70.1	73.0	73.2	7.6	12.8	1.3	0.12
	Places365	81.8	80.1	83.4	6.1	10.1	0.5	0.14
	CelebA	92.2	94.8	95.8	10.2	44.6	11.7	-0.06
	upside-down ID	50.5	95.5	80.4	3.8	69.3	1.5	-0.43
FMNIST	MNIST	100.0	100.0	100.0	0.0	0.0	100.0	0.30
	KMNIST	99.9	100.0	100.0	0.0	0.0	100.0	0.04
	Omniglot	99.8	100.0	99.9	0.0	1.5	98.5	0.38
	notMNIST	100.0	99.4	100.0	5.3	0.0	94.7	0.33
	upside-down ID	96.7	100.0	99.7	0.0	21.1	78.9	0.20

S5 All Pairs

Table S5: AUROC (%) of every pair (final evaluation; Upright kNN, tuned kNN and CFS averaged over three seeds). Best prior: the best of the 11 prior methods on that pair (chosen after the fact). Difference: Upright kNN minus that method; **bold** = win, *italic* = loss, plain = tie (bootstrap rule of the paper).

ID	OOD	Ours	best prior		diff.	tuned kNN	diff.
CIFAR-10	CIFAR-100	80.3	78.4	Rot. pred.	+1.9	77.1	+**3.2**
	SVHN	99.9	99.5	DiffPath	+**0.3**	99.7	+0.1
	CelebA	96.0	86.2	Rot. pred.	+**9.8**	89.6	+**6.4**
	Textures	99.5	97.5	CFS-kNN	+**2.0**	99.6	−0.1
	Tiny-ImageNet	96.1	83.7	Rot. pred.	+**12.4**	92.1	+**3.9**
	LSUN	96.7	85.0	IC-PNG	+**11.7**	96.2	+0.6
	ImageNet (resized)	97.6	88.1	CFS-kNN	+**9.5**	97.1	+0.5
	ImageNet32	96.3	89.5	CFS-kNN	+**6.9**	96.2	+0.1
	Places365	96.0	84.7	IC-PNG	+**11.3**	95.7	+0.3
	upside-down ID	96.1	99.3	Rot. pred.	*−3.2*	77.2	+**18.9**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	CFS	0.0	100.0	0.0
SVHN	CIFAR-10	99.6	99.0	DiffPath	+**0.7**	99.2	+**0.5**
	CIFAR-100	99.7	99.2	DiffPath	+**0.5**	99.4	+**0.3**
	CelebA	99.2	100.0	NLL	*−0.8*	98.2	+**1.0**
	Textures	99.9	99.9	MSMA*	0.0	99.8	+**0.1**
	Tiny-ImageNet	99.7	98.8	DiffPath	+**0.9**	99.4	+**0.4**
	LSUN	99.9	99.9	DiffPath	0.0	99.7	+**0.2**
	ImageNet (resized)	99.9	99.8	DiffPath	+0.1	99.7	+**0.2**
	ImageNet32	99.9	100.0	DiffPath	−0.1	99.7	+**0.2**
	Places365	99.8	99.8	DiffPath	0.0	99.7	+**0.2**
	upside-down ID	64.5	77.7	Rot. pred.	*−13.1*	55.9	+**8.6**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	CFS	0.0	100.0	0.0
CelebA	CIFAR-10	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	CIFAR-100	100.0	100.0	Rot. pred.	0.0	99.9	+**0.1**
	SVHN	100.0	100.0	DiffPath	0.0	100.0	0.0
	Textures	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	Tiny-ImageNet	100.0	99.9	DiffPath	+**0.1**	99.9	0.0

ID	OOD	Ours	best prior		diff.	tuned kNN	diff.
	LSUN	100.0	100.0	Rot. pred.	0.0	99.9	0.0
	ImageNet (resized)	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	ImageNet32	100.0	100.0	Rot. pred.	0.0	99.9	+**0.1**
	Places365	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	upside-down ID	100.0	100.0	Rot. pred.	0.0	99.8	+**0.2**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	IC-PNG	0.0	100.0	0.0
CIFAR-100	CIFAR-10	64.1	65.9	CFS-kNN	−1.8	66.6	*−2.5*
	SVHN	99.5	97.8	DiffPath	+**1.7**	98.9	+**0.6**
	CelebA	78.4	61.3	IC-J2K	+**17.1**	77.7	+0.7
	Textures	99.0	93.3	CFS	+**5.7**	98.8	+0.2
	Tiny-ImageNet	86.4	70.4	DiffPath	+**16.0**	82.6	+**3.8**
	LSUN	81.7	78.5	IC-PNG	+**3.1**	82.4	−0.7
	ImageNet (resized)	88.3	73.0	CFS-kNN	+**15.3**	88.1	+0.2
	ImageNet32	88.9	81.6	CFS-kNN	+**7.3**	89.4	−0.5
	Places365	83.0	79.0	IC-PNG	+**4.0**	83.7	−0.7
	upside-down ID	79.1	96.5	Rot. pred.	*−17.4*	56.8	+**22.4**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	MSMA*	0.0	100.0	0.0
ImageNet-64	SSB-hard	67.3	56.2	MSMA*	+**11.2**	67.8	−0.5
	NINCO	71.7	65.0	Rot. pred.	+**6.7**	68.3	+**3.4**
	iNaturalist	71.9	84.4	NLL	*−12.5*	68.1	+**3.8**
	Textures	93.6	92.0	CFS-kNN	+1.6	91.0	+**2.6**
	OpenImage-O	73.2	69.6	CFS-kNN	+**3.6**	70.1	+**3.1**
	Places365	83.4	74.8	MSMA*	+**8.7**	81.8	+**1.6**
	CelebA	95.8	76.5	MSMA*	+**19.3**	92.2	+**3.6**
	upside-down ID	80.4	94.5	Rot. pred.	*−14.1*	50.5	+**29.9**
FashionMNIST	MNIST	100.0	100.0	CFS-kNN	0.0	100.0	0.0
	KMNIST	100.0	100.0	CFS-kNN	0.0	99.9	+**0.1**
	Omniglot	99.9	99.9	Rot. pred.	0.0	99.6	+**0.3**
	notMNIST	100.0	100.0	CFS-kNN	0.0	99.9	0.0
	upside-down ID	99.7	99.9	Rot. pred.	−0.2	96.0	+**3.7**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	IC-PNG	0.0	100.0	0.0

S6 FPR95

Method	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip
Upright kNN	19.4	1.5	0.0	62.0	75.7	0.1	26.5	41.3
kNN (tuned)	28.8	1.7	0.0	63.3	84.5	0.4	29.8	64.2
kNN (basic)	54.3	8.9	0.0	69.0	88.6	1.3	37.0	71.9
NLL	86.7	3.4	41.0	91.3	88.9	98.8	68.3	90.4
IC-PNG	75.6	31.2	44.0	84.5	89.4	67.8	65.4	93.2
IC-J2K	72.3	52.5	53.4	85.3	93.5	95.8	75.5	92.6
TT	90.0	3.8	41.6	92.9	89.7	86.5	67.4	91.8
MSMA*	84.9	4.6	1.1	83.5	89.9	22.4	47.7	77.2
DiffPath	72.3	1.6	0.3	79.2	91.5	72.8	53.0	78.4
GEPC	90.3	51.1	4.2	90.0	92.9	77.2	67.6	95.8
EigenScore	85.5	61.0	55.6	94.3	95.6	27.5	69.9	74.4
CFS	73.5	56.2	0.3	78.9	85.9	13.3	51.4	86.5
CFS-kNN	58.9	41.1	0.0	79.2	85.7	0.2	44.2	65.3
Rot. pred.	58.2	27.0	0.0	82.3	91.9	1.4	43.5	19.8

Table S6: FPR at 95% TPR (%, lower is better; final evaluation), mean over the natural OOD sets of each ID dataset, their mean, and the mean over the upside-down sets.

S7 Ablations

Variant	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip	Selection
Upright kNN, Eq. (5)	95.9	99.4	100.0	84.6	80.3	99.9	93.4	86.4	93.3
kNN term only	93.9	99.1	100.0	83.5	78.1	99.9	92.4	72.7	93.0
orientation term only	94.2	95.8	100.0	72.1	77.2	99.7	89.8	94.7	87.6
no temperature ($T = 1$)	95.9	99.3	100.0	84.6	80.3	99.9	93.3	87.7	93.2
Fisher (sum of log p-values)	96.5	99.4	100.0	82.3	80.6	100.0	93.1	92.3	92.5
rotated bank for $\bar{p}$ ($4R$, d re-estimated)	91.8	99.4	99.9	84.0	80.5	99.8	92.6	86.5	92.9
head trained with rotated noise*	94.5	99.5	100.0	83.6	80.1	99.9	92.9	78.8	93.1
orientation term over 4 test views*	95.3	99.1	100.0	84.3	79.0	99.9	92.9	88.1	93.2

Table S7: Development evaluation images, noise seed 0: mean AUROC over the natural OOD sets, the upside-down mean, and the mean on the selection data. No variant raises the selection-data mean, so within this table the frozen configuration would also be chosen on the selection data alone. *Approximation from stored features whose noise was rotated with the image.

S8 Mirror Test

	vertical M by depth				horizontal M (control)				kNN term of ours			
ID	1	1/2	1/4	1/8	1	1/2	1/4	1/8	vert.	horiz.	ρ	ρ_{H}
CIFAR-10	52.1	67.3	87.8	–	50.0	52.8	54.2	–	77.7	52.4	0.91(29)	0.61
SVHN	50.4	51.4	61.9	65.3	50.2	52.0	64.1	69.1	55.7	57.2	−0.53(13)	−0.56
CelebA	58.8	97.9	99.8	97.1	49.6	50.3	51.0	51.1	99.4	50.3	0.91(33)	0.46
CIFAR-100	51.4	56.3	63.3	–	50.2	52.2	55.7	–	56.6	52.6	0.68(29)	0.68
ImageNet-64	49.9	51.1	53.8	62.1	50.0	50.1	49.4	49.9	50.2	49.9	0.89(30)	−0.14
FashionMNIST	69.4	93.7	98.2	95.8	59.1	72.4	80.6	77.4	96.6	77.0	0.87(22)	0.84

Table S8: Mirror test M (%, development images, seed 0, σ_1) of the kNN term ($R = 1$) of the last decoder block at each depth (resolution ratio; –: no such level), with the horizontal flip as control; the kNN term of Upright kNN (three blocks, frozen R); and the Spearman correlation over all blocks (number in parentheses) between the vertical M and the natural-OOD AUROC of the block (ρ), and the same for the horizontal M (ρ_{H}).

S9 ImageNet-1K at 64×64 by Set

Method	SSB-hard	NINCO	iNat	Textures	OI-O	V2	C	R	FPR near	FPR far	W/T/L
Upright kNN	68.6	71.9	70.1	93.2	73.9	52.4	92.3	77.9	83.9	74.4	–
kNN term	69.7	69.3	66.2	91.4	71.5	52.2	92.1	73.4	82.2	76.6	4/0/1
orient. term	56.4	70.6	74.4	86.1	71.9	50.9	80.7	77.8	91.4	81.0	2/2/1
kNN (basic)	56.0	65.0	46.8	89.9	67.5	50.6	83.6	63.9	87.8	74.2	5/0/0
CFS (diag.)	57.5	63.8	68.5	88.7	67.6	52.0	77.3	68.7	86.0	76.0	4/1/0
CFS (kNN)	52.3	60.4	53.4	90.3	69.4	51.5	80.7	67.0	92.4	71.2	5/0/0
Rot. pred.	43.8	62.4	67.7	78.5	64.9	49.7	63.2	65.1	93.2	89.4	4/1/0
NLL	49.1	51.8	83.8	52.9	58.2	51.0	43.3	31.3	93.2	79.0	4/0/1
IC-PNG	49.8	55.7	65.5	47.8	57.9	48.1	56.6	45.7	92.4	87.5	5/0/0
IC-J2K	49.0	53.4	65.0	46.5	55.6	48.3	59.9	36.1	93.5	91.8	5/0/0
TT	53.6	59.8	74.0	54.7	61.5	49.6	66.1	62.7	90.2	89.9	4/0/1
MSMA*	56.8	56.7	41.2	81.5	56.7	46.9	79.1	72.1	90.3	82.3	5/0/0
DiffPath	50.4	56.3	59.4	70.4	58.0	52.1	57.6	69.6	91.6	87.5	5/0/0
GEPC	46.3	53.7	46.2	65.4	57.7	49.8	57.8	59.0	94.1	87.2	5/0/0
EigenScore	47.1	45.6	42.3	48.9	46.9	49.5	54.0	67.8	96.8	95.1	5/0/0

Table S9: OpenOOD v1.5 ImageNet-1K, 64×64, EDM2-XS (AUROC and mean FPR95, %; 1,000 common images per set). W/T/L of Upright kNN over the five OOD sets.

Method	1	2	3	4	5
Upright kNN	88.4	91.3	91.6	93.3	94.5
kNN term	88.9	91.1	91.5	92.9	93.7
orient. term	74.9	78.6	79.5	82.0	84.0
kNN (basic)	75.3	81.9	82.5	86.8	88.3
CFS (diag.)	65.2	72.1	76.2	81.3	83.9

Table S10: AUROC (%) of ImageNet val against ImageNet-C by corruption severity (64×64, full sets, three seeds). Density-based scores flag corrupted images more as the severity grows.

S10 ImageNet at 256×256 without Resolution Control

Method	SSB-hard	NINCO	iNat	Textures	OI-O	Flip
Upright kNN	93.7	80.6	84.1	78.5	54.0	78.5
kNN term	94.9	81.4	86.2	75.1	48.6	50.6
orient. term	75.3	66.1	61.3	77.2	71.5	92.6
kNN (basic)	78.5	66.1	85.6	50.3	13.0	50.3
CFS (diag.)	85.4	62.5	57.3	88.5	76.5	49.4
CFS (kNN)	82.8	65.9	57.9	82.1	41.6	50.7
MSMA*	63.7	62.0	33.9	88.2	79.5	50.0
DiffPath	62.6	53.7	39.3	60.6	44.9	50.8
GEPC	70.6	54.0	48.4	49.1	40.0	50.8

Table S11: ADM 256×256 with each set's images cropped from their stored resolution (500 common images per set). SSB-hard is stored at 224 pixels (short side, all 200 sampled images) and is upsampled, whereas ImageNet validation has a median short side of 375 pixels (3% below 256); ImageNet-C is also stored at 224 pixels. Every feature-based detector separates SSB-hard far better than at 64×64, a resolution shortcut; the paper therefore reports this setting without SSB-hard and ImageNet-C.

S11 Fusions

Score	Mean	Flip	W/T/L of ours
Upright kNN (sum, Eq. 2)	**93.4**	86.6	–
kNN term only	92.5	72.8	28/18/1
orientation term only	89.9	94.7	29/14/4
rotation classifier (ResNet-18)	84.0	94.6	35/12/0
kNN + rot. classifier, z-sum	91.2	92.4	30/17/0
kNN + rot. classifier, Fisher	91.2	92.9	19/26/2
kNN + orient. term, z-sum	93.1	92.4	23/17/7
kNN + orient. term, Fisher	93.1	92.5	13/26/8
Upright kNN with $d = 1$	92.0	94.2	24/20/3

Table S12: Principled sum vs. naive fusions (final evaluation, AUROC, %; Mean and Flip as in Table 2 of the paper). z-sum: sum of z-scores; Fisher: sum of log empirical ID p-values; both normalise each score with additional ID images (the development split), which Upright kNN does not need. W/T/L over the natural-OOD pairs. This comparison was added after the frozen evaluation and reuses its images.

S12 CSI

CSI [Tack *et al.*, 2020] is run with the official code (commit 60742b6, unmodified) in the unlabeled multi-class setting of its repository: ResNet-18, simclr_CSI with rotations as shifting transformations, 1,000 epochs, LARS with learning rate 0.1, 10 warm-up epochs, cosine schedule, weight decay 10^{-6}, temperature 0.5, on one GPU with a batch of 128 images (the official total batch of 512 rotated images). The training images are the bank images of each 32×32 ID dataset, i.e. the images of the kNN bank and of the rotation head of Upright kNN. The score is the official CSI score with the repository's evaluation flags (10 augmented samples, crop scale 0.54). A thin wrapper feeds our arrays and replaces two packages that do not build with our PyTorch version (a re-implementation of the LARS optimizer of torchlars 0.1.2 and a stub of the multi-GPU gather, which a single-GPU run does not call). The protocol was written before training. [Results, training cost and the reproduction check against the published CIFAR-10 → CIFAR-100 AUROC (89.3%) will be added when the runs finish.]

S13 EuroSAT

Method	C10	C100	SVHN	CelebA	Tex.	TIN	LSUN	IN	IN32	P365	Mean	Flip	Rot. 90
Upright kNN	100.0	100.0	99.3	100.0	99.5	99.9	100.0	99.9	100.0	100.0	99.8	64.0	75.8
kNN term	99.8	99.8	99.7	99.9	99.6	99.8	99.8	99.8	99.8	99.8	99.8	54.2	55.1
orient. term	99.3	98.9	81.8	99.7	91.4	97.3	99.2	96.9	98.6	98.1	96.1	75.6	91.1
kNN (basic)	99.6	99.7	99.8	99.8	99.6	99.6	99.6	99.6	99.7	99.6	99.7	52.8	54.0

Table S13: EuroSAT as ID (U-Net trained on EuroSAT, $R = 5$ fixed in advance, three seeds): AUROC (%) against ten natural OOD sets (500 images each), and of ID images against their vertical flips and 90° rotations.

S14 Compute and Reproducibility

All features were extracted on NVIDIA RTX 2080 Ti GPUs (11 GB). Offline cost of Upright kNN: one pass of the denoiser over the four rotations of the bank (about 7 minutes for 50,000 CIFAR-10 images on one GPU) and about 10 s of head training. Test cost: one denoiser pass, inner products with the bank, and the pooled grid statistics and MLP (about 2 ms on a CPU). All analysis scripts read per-image scores and write every number and table of the paper and of this supplement; the code, the protocol documents and the per-image scores will be released with the camera-ready version.

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